Protocols, cryptanalysis and mathematical cryptology (INFO-F514)

Cryptanalysis

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Sensitization campaign

Together against sexual harassment and violences
https://www.ulb.be/fr/sante-et-bien-etre/si-cest-pas-oui-cest-non

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Cryptography vs Cryptanalysis

- Cryptography is the science and art of ensuring private and authenticated communications
- Cryptanalysis is the science and art of breaking it!

- Cryptanalysis is useful for cryptography
  - Cryptanalysis attempts give confidence on hardness
  - Cryptanalysis benchmarks help determining key sizes
  - Cryptanalysis successes teach us design lessons

  Good criminals make good policemen too !

Outline

Introduction

The birthday paradox

Solving discrete logarithm problems

Discrete logarithm variants and factoring

Conclusion

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Remember: "Provable Security"

- Precisely define what it means to break the protocol
  - Adversary's goal
  - Adversary's resources
  - Adversary's access to the system
- Choose your favorite hard problem
  - A computational problem that cannot be solved,
    even by clever people with the best computers available
- Build a protocol so that you can prove

  Breaking the protocol $\Rightarrow$ Solving the hard problem

Limits in Provable Security

- Remember "provable security"

  Breaking the protocol $\Rightarrow$ Solving the hard problem

- How can this go wrong?
  - Find out that security proof is wrong
  - Solve the underlying hard problem
  - Find weaknesses not covered by the proof

Why solving hard problems?

- Remember "provable security"

  Breaking the protocol $\Rightarrow$ Solving the hard problem

- Proof only useful if the problem is indeed "hard"
- Can often improve security at the price of efficiency
- Need cryptanalysis benchmark to select the key sizes optimizing security and efficiency

How to solve hard problems?

- Optimize existing algorithms and design new ones
- Implement algorithms on all sorts of computing devices
- Benchmark cryptanalysis results
- Try and engage more people to try!

A glance at a cryptanalyst's toolbox

- Computational number theory
- Birthday paradox attacks
- Linear and differential cryptanalysis
- Generic discrete logarithm algorithms
- Index calculus algorithms
- Lattice reduction algorithms
- Groebner basis algorithms, SAT solvers,...
- Anything that can help!

A glance at a cryptanalyst's toolbox (2)

- Laptops and desktops
- Computing servers
- FPGAs
- Dedicated hardware
- Sony Playstations
- Cloud computing
- Mobile phones
- Anything that can help!

Playstation games for cryptanalysts

- Advanced attacks on MD5 led to Nostradamus attack

Public challenges

- Goal: encourage cryptanalysis attempts to estimate concrete hardness of a problem, and parameters to be used in practice
  - RSA factoring challenges (ended 2007)
  - Certicom ECDLP challenges (solved up to 109 bits)
  - Darmstadt lattice challenges
  - Fukuoka MQ challenges
- Contribute and enter "Hall of Fame" of any of them!

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Birthday paradox

- Suppose there are N_2 people in a room. What is the probability that two people have the same birthday?
- How many people needed to have a probability larger than 50% ?
- Answer is **23**:

$$\Pr[\text{all distinct}] = 1 \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \ldots \cdot \frac{365-22}{365} < \frac{1}{2}$$

Birthday paradox

- Suppose you choose N_2 elements randomly in a set of N elements. What is the probability that two elements are equal?
- How should N_2 be wrt N to have a probability larger than 50% ?
- Answer is $O(\sqrt{N})$:

$$\begin{aligned}
\Pr[\text{all distinct}] &= 1 \cdot \frac{N-1}{N} \cdot \frac{N-2}{N} \cdot \ldots \cdot \frac{N-N_2+1}{N} \\
&\approx e^{-\frac{1}{N}} \cdot e^{-\frac{2}{N}} \cdot \ldots \cdot e^{-\frac{N_2-1}{N}} \\
&\approx e^{-\frac{N_2(N_2-1)}{N}}
\end{aligned}$$

Taking $N_2 \approx \sqrt{N}$ ensures $1 - \Pr[\text{all distinct}] \in \Omega(1)$

Computing collisions

- Remember: a hash function $H : \{0,1\}^* \rightarrow \{0,1\}^n$ is collision resistant if it is hard to find (m, m') with $H(m) = H(m')$ but $m \neq m'$
- Simple algorithm: choose random messages m_i and compute $h_i = H(m_i)$ until you get a collision
- By birthday paradox, you only need $O(2^{n/2})$ elements to get a collision

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Discrete logarithm algorithms

- Remember discrete logarithm problem (DLP): given a finite cyclic group G of prime order p, a generator g and a second element h, find k such that $h = g^k$
- Hardness depends on the group
- Pollard's rho, Baby-Step-Giant-Step algorithms work for any group, with complexity $O(\sqrt{|G|})$
- These are current best for well-chosen elliptic curves
- Index calculus algorithms have complexity $\leq L_{|G|}(1/3, c)$ when G is multiplicative group of finite field
- Much better complexity when characteristic is small (quasi-polynomial time algorithm)

Exhaustive search

- Let $h = g^k$. You want to compute k.
- Exhaustive search algorithm

```
1: k ← 1; h' ← g
2: if h' = h then
3:     return k
4: else
5:     k ← k + 1; h' ← h'g
6:     Go to Step 2
7: end if
```

- Generic algorithm
- Time complexity $|G|$ in the worst case, $|G|/2$ on average
- Can we do better?

Baby step, giant step (BSGS)

- Let $h = g^k$. You want to compute k.
- Let $N' = \lceil \sqrt{|G|} \rceil$
- There exist $0 \leq i, j < N'$ such that $k = jN' + i$

$$h = g^{jN'+i} \Leftrightarrow hg^{-jN'} = g^i$$

- Compute $L_B := \{g^i | i = 0, \ldots, N' - 1\}$
- Compute $L_G := \{hg^{-jN'} | j = 0, \ldots, N' - 1\}$
- Find an element in $L_G \cap L_B$
- Attack requires time and memory $O(\sqrt{|G|})$

Pollard's rho (iterative function)

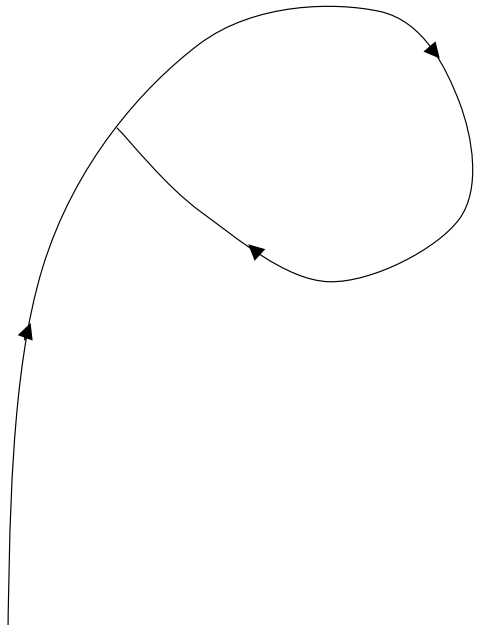
- Define G_1, G_2, G_3 of about the same size such that $G = G_1 \cup G_2 \cup G_3$ and $G_i \cap G_j = \{\}$
- Over $\mathbb{Z}_p^*$, can choose
  $G_1 = \{0, \ldots, \lfloor p/3 \rfloor\}$,
  $G_2 = \{\lfloor p/3 \rfloor + 1, \ldots, \lfloor 2p/3 \rfloor\}$,
  $G_3 = \{\lfloor 2p/3 \rfloor + 1, \ldots, p-2\}$
- Define a function $f : G \to G$ such that

$$\begin{cases} f(z) = zg & z \in G_1 \\ f(z) = z^2 & z \in G_2 \\ f(z) = zh & z \in G_3 \end{cases}$$

(original definition, but other definitions possible)

Pollard's rho (intuition)

- Start from $g_0 := g$ and apply f recursively to get g_i
- If f is "random enough", we obtain random elements in G and a collision after $O(\sqrt{|G|})$ elements
- By the way f is defined, we can keep track of a_i, b_i such that $g_i = g^{a_i} h^{b_i}$
- Collision gives DLP solution



Pollard's rho (simplest version)

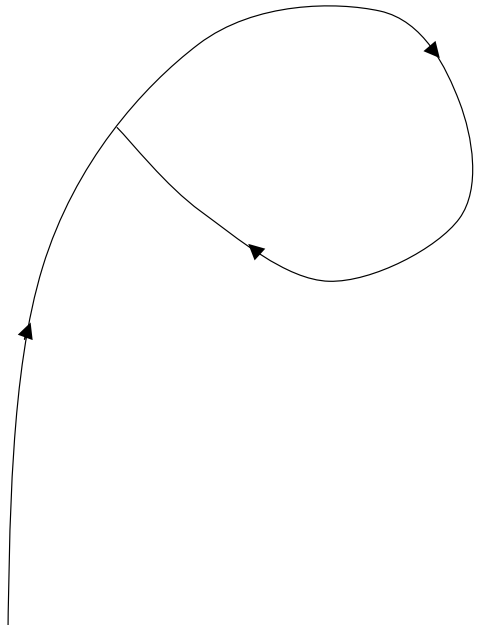
$N \leftarrow \lceil \sqrt{|G|} \rceil$
$a \leftarrow 1; b \leftarrow 0; \tilde{h} \leftarrow g; L \leftarrow \{(a, b, \tilde{h})\}$
for $k \in \{2, \ldots, N\}$ **do**
    if $\tilde{h} \in G_1$ **then** $a \leftarrow a + 1; \tilde{h} \leftarrow \tilde{h}g$
    if $\tilde{h} \in G_2$ **then** $a \leftarrow 2a; b \leftarrow 2b; \tilde{h} \leftarrow (\tilde{h})^2$
    if $\tilde{h} \in G_3$ **then** $b \leftarrow b + 1; \tilde{h} \leftarrow \tilde{h}h$
    $L \leftarrow L \cup \{(a, b, \tilde{h})\}$
end for
Find distinct $(a_i, b_i, \tilde{h}) \in L$, $i = 1, 2$
if no such elements **then abort**
return $-(a_1 - a_2)/(b_1 - b_2) \bmod |G|$

Pollard's rho analysis

- Correctness:
  - Every $(a, b, \tilde{h})$ in the list satisfies $\tilde{h} = g^a h^b$
  - $g^{a_1} h^{b_1} = g^{a_2} h^{b_2}$ implies $h = g^{-\frac{a_1 - a_2}{b_1 - b_2}}$
- Time and memory costs $O(N) = O(\sqrt{|G|})$
- Good probability of success by birthday's paradox

Pollard's rho (improvement)

- Let $(L_1, L_1 + L_2)$ be the indices of first collision
- Then $(L_1 + j, L_1 + j + kL_2)$ also collide
- For j, k such that $L_1 + j = kL_2$, we have $L_1 + kL_2 + j = 2(L_1 + j)$
- Now search for $(a_i, b_i, \tilde{h}_i)$ and $(a_{2i}, b_{2i}, \tilde{h}_{2i})$ such that $\tilde{h}_i = \tilde{h}_{2i}$
- Only requires constant size memory



Pohlig-Hellman

- Assume $|G| = n_1 n_2$ and let g a generator of G
- $h = g^k$ implies $h^{n_1} = (g^{n_1})^k$
  where g^{n_1} generates a subgroup of order n_2
- Solving DLP in that subgroup gives k mod n_2
- Repeat for each (power of prime) factor of $|G|$
- Use a constructive version of Chinese Remainder Theorem (extended Euclide algorithm) to get k (see next slides)
- Countermeasure: use a large prime order group

Pohlig-Hellman: example

- Let $p := 11$
- Then 2 is a generator of $\mathbb{F}_p^*$
  - Successive powers are 2,4,8,5,10,9,7,3,6,1
  - Order is $p - 1 = 2 \cdot 5$
- Let $g = 2$ and $h = 7$
  - $g^5 = 10$, $h^5 = 10$ hence $x = 1 \bmod 2$
  - $g^2 = 4$, $h^2 = 5$ hence $x = 2 \bmod 5$
  - Deduce $x = 7$
- Let $g = 2$ and $h = 3$
  - $g^5 = 10$, $h^5 = 1$ hence $x = 0 \bmod 2$
  - $g^2 = 4$, $h^2 = 9$ hence $x = 3 \bmod 5$
  - Deduce $x = 8$

Chinese Remainder Theorem

- Let p_1, p_2 prime and let $n = p_1 p_2$. For any x_1, x_2, there is one and only one value of x modulo n such that $x = x_1 \bmod p_1$ and $x = x_2 \bmod p_2$
- This value can be efficiently computed
  1. Compute $(p_1^{-1} \bmod p_2)$ and $(p_2^{-1} \bmod p_1)$ using the Extended Euclide algorithm (see next slide)

     $$1 = p_1(p_1^{-1} \bmod p_2) + p_2(p_2^{-1} \bmod p_1)$$

  2. Deduce

     $$x = x_2 p_1(p_1^{-1} \bmod p_2) + x_1 p_2(p_2^{-1} \bmod p_1)$$

Extended Euclide algorithm

- Goal: compute d, r, s such that $d = ra + sb = \gcd(a, b)$

  Require: $a \geq b$

  Ensure: $d = \gcd(a, b)$ **and** r, s, such that $ar + bs = d$

  if $b|a$ **then**
   return $a, 0, 1$
  else
   Compute q such that $0 < a - qb < b$
   $d, r, s \leftarrow \gcd(b, a - qb)$
   return $d, s, r - qs$
  end if

- Indeed if $rb + s(a - qb) = d$ then $sa + (r - qb)b = d$
- Complexity $O(|a|^2)$; best algorithms achieve $O(|a| \log |a|)$

Extended Euclide algorithm: example

- Consider $a = 31$, $b = 11$
- We have

$$\begin{aligned} 31 &= 2 \cdot 11 + 9 \\ 11 &= 1 \cdot 9 + 2 \\ 9 &= 4 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 \end{aligned}$$

- Hence

$$\begin{aligned} 9 &= 1 \cdot 31 - 2 \cdot 11 \\ 2 &= 1 \cdot 11 - 1 \cdot 9 = -1 \cdot 31 + 3 \cdot 11 \\ 1 &= 1 \cdot 9 - 4 \cdot 2 = 5 \cdot 31 - 14 \cdot 11 \end{aligned}$$

Chinese Remainder Theorem (2)

- More generally if $n = \prod_{i=1}^{N} p_i^{e_i}$, for any $x_1, x_2, \ldots, x_N$, there is one and only one value of x modulo n such that $x = x_i \bmod p_i^{e_i}$
- This value can be efficiently computed

Generic DLP algorithms

- Exhaustive search, Pollard's rho, Baby-step-giant-step and Pohlig-Hellman algorithms work **for any group**
- For any group G, solving DLP in G costs at most $O(\sqrt{q})$ group operations, where q is largest prime dividing $|G|$
- Countermeasure: take a large prime order group

DLP algorithms for finite fields

- In cryptography we often use multiplicative groups of finite fields
- For $\mathbb{F}_p^*$ the best algorithm has complexity[1]

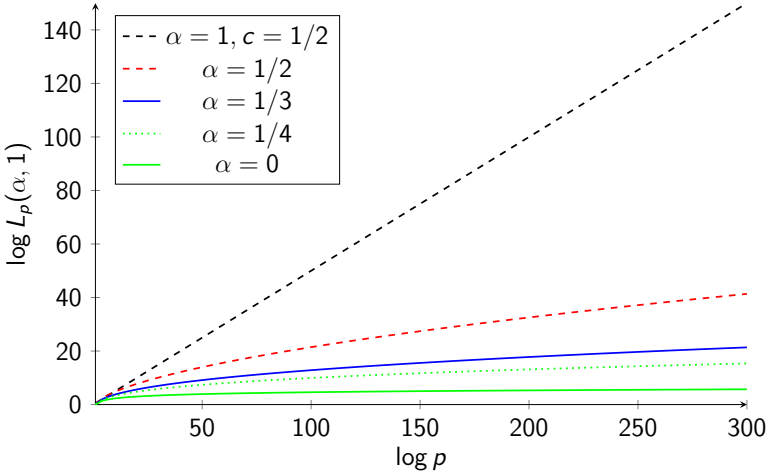
$$L_p\left(\frac{1}{3}, \sqrt[3]{\frac{64}{9}}\right) = \exp\left(\left(\sqrt[3]{\frac{64}{9}} + o(1)\right)(\log p)^{1/3}(\log\log p)^{2/3}\right)$$

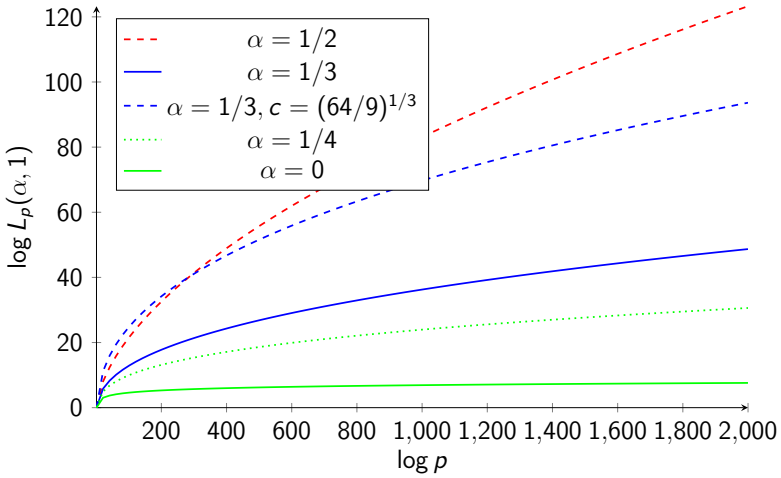
- Current record has 795 bits (Dec 2019)
- For $\mathbb{F}_{2^n}^*$ best algorithm has quasi-polynomial complexity since 2013, broken in theory and in practice

[1] $L_p(\alpha, c) = \exp((c + o(1))(\log p)^{\alpha}(\log\log p)^{1-\alpha})$

Subexponential L function

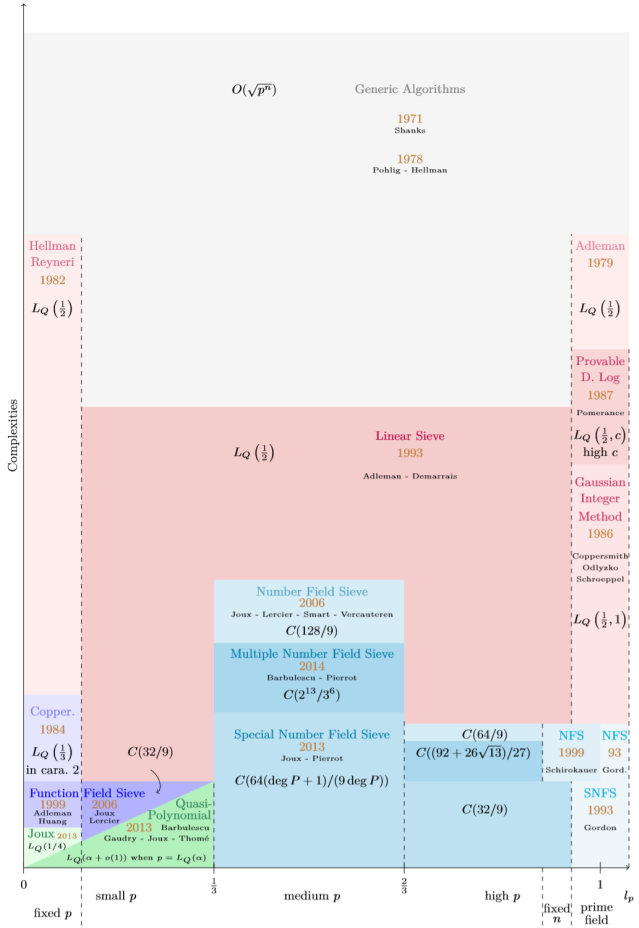
$$L_p(\alpha, c) = \exp((c + o(1))(\log p)^{\alpha}(\log\log p)^{1-\alpha})$$





Diffie-Hellman paper initially suggested 200-bit primes

DLP algorithms for finite fields



Picture credit: Joux-Odlyzko-Pierrot

Index calculus

- Generic framework to solve discrete logarithm problems, but some steps are group-specific
- Let g, h a DLP problem
- Define a *factor basis* $\mathcal{F} \subset G$, ensuring $\mathcal{F}$ contains a generator (most elements in G are generators)
- Can assume $g \in \mathcal{F}$, otherwise do the following:
  - Pick a generator $g' \in \mathcal{F}$
  - Compute a such that $g = (g')^a$
  - Compute b such that $h = (g')^b$
  - Compute $k = b/a \bmod |G|$
- Remark: size of $\mathcal{F}$ will be optimized for efficiency

Index calculus

- Find about $|\mathcal{F}|$ *relations* between factor basis elements

$$\mathcal{R}_j : \quad \prod_{f_i \in \mathcal{F}} f_i^{a_{i,j}} = 1$$

(the algorithm to compute the relations is group-specific)

- Deduce

$$\sum_{f_i \in \mathcal{F}} a_{i,j} \log_g f_i = 0$$

or

$$\begin{pmatrix} a_{1,1} & \dots & a_{|\mathcal{F}|,1} \\ \vdots & & \vdots \\ a_{1,|\mathcal{F}|} & \dots & a_{|\mathcal{F}|,|\mathcal{F}|} \end{pmatrix} \begin{pmatrix} \log_g f_1 \\ \vdots \\ \log_g f_{|\mathcal{F}|} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Index calculus

- Use linear algebra to compute all $\log_g f_i$,
  the discrete logarithms of factor basis elements
- Deduce the discrete logarithm of h
  (this part is group-specific and may involve several steps)
- Remarks:
  - Relations often involve few elements,
    hence linear algebra is sparse
  - In some cases, h is included in the factor basis and the last step is avoided: linear algebra produces $\log_g h$

Example: a naive index calculus for $\mathbb{F}_p^*$

- DLP: given $g, h \in \mathbb{F}_p^*$, find k such that $h = g^k$
- Factor basis made of **small primes** and h

$$\mathcal{F}_B := \{\text{primes } p_i \leq B\} \cup \{h\}$$

- **Relation search**
  - Compute $r_j := g^{a_j} h^{b_j}$ for random $a_j, b_j \in \{1, \ldots, p-1\}$
  - **If** all factors of r_j are $\leq B$, we have one relation

$$g^{a_j} h^{b_j} = \prod_{p_i \in \mathcal{F}} p_i^{e_{i,j}}$$

- **Linear algebra** produces a, b such that $g^a h^b = 1$

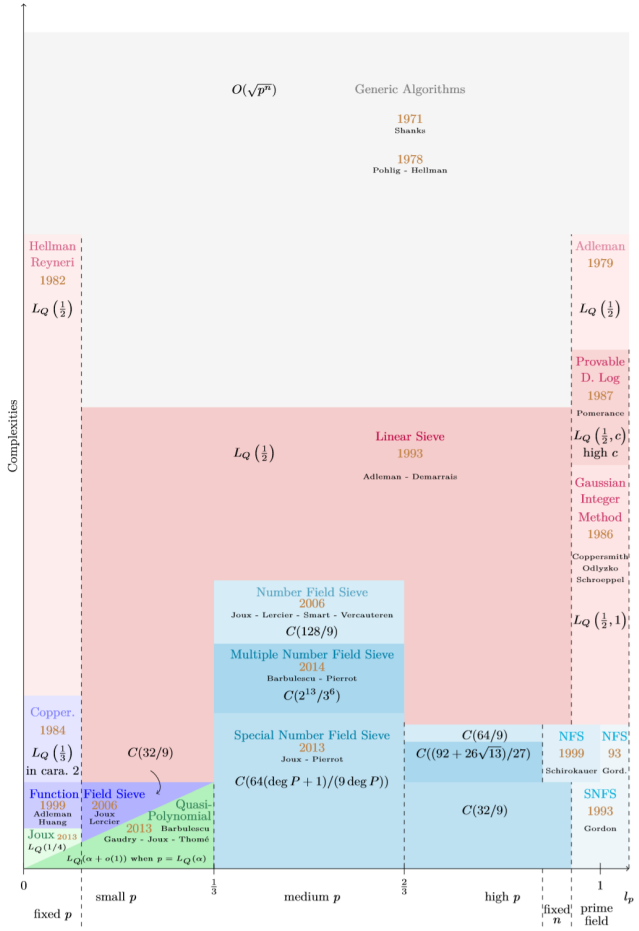
Naive analysis of naive index calculus

- Choose $\log B \approx (\log p)^{1/2}$
- $|\mathcal{F}| \approx B/\log B \approx 2^{(\log p)^{1/2}-(\log\log p)^{-1/2}} \approx 2^{(\log p)^{1/2}}$
- B-smoothness probability
  - $u = \log p/\log B \approx (\log p)^{1/2}$
  - $\rho(u) = (\log p)^{-1/2(\log p)^{1/2}} \approx 2^{-1/2(\log p)^{1/2}(\log\log p)}$
- Number of random trials to get $|\mathcal{F}|$ relations is

$$\approx |\mathcal{F}|\rho(u)^{-1} \approx 2^{(1/2+o(1))(\log p)^{1/2}(\log\log p)}$$

- Each trial has polytime complexity in $\log p$
- Linear algebra cost is $|\mathcal{F}|^{\omega} \approx 2^{\omega(\log p)^{1/2}}$
- Total cost dominated by relation search
- $B \approx L_p(1/2; c)$ leads to slighly better cost $L_p(1/2; c')$

DLP algorithms for finite fields



ECDLP: state-of-the-art

- We have an $L(2/3)$ algorithm to solve ECDLP over fields $\mathbb{F}_{q^n}$ if q and n have the right size
- In most applications we are interested in ECDLP over either prime fields, or $\mathbb{F}_{2^n}$ with extension degree n prime
- Some algorithms have been suggested in those cases, but their complexity is unknown

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Discrete logarithms and related problems

- Let G be a finite cyclic group of prime order q, and let g be a generator
- Discrete logarithm problem: given G, g and h, find x such that $h = g^x$
- Diffie-Hellman problem: given G, g, g^a, g^b, compute g^{ab}
- Decisional Diffie-Hellman problem: let G, g, g^a, g^b, g^c where c is either ab mod q or random (with equal probability), guess correct distribution with probability significantly bigger than $1/2$

- Solving discrete logarithm is a natural approach to solving CDH or DDH, but there could a priori be other methods

DLP vs CDH

- CDH is *roughly* equivalent to DLP
  - Given CDH solver and some precomputation (feasible for the groups used in practice), can solve DLP
- Sketch:
  - Using oracle $CDH(g^a, g^b) = g^{ab}$, can build $CDH(g^x, g^x) = g^{x^2}$, then $CDH(g^{x^2}, g^{x^2}) = g^{x^4}, \ldots$
  - Can build g^{x^e} with "square-and-multiply in exponent" for any e
  - If $|G| - 1$ is smooth, fix g_0 and use Pohlig-Hellman to find x_0 such that $g_0^{g^{x_0}} = g^x$ ie solve DLP in exponent
  - In general: fiind smooth order elliptic curve modulo $|G| - 1$, with x the x-coordinate of a point
- See Maurer *Towards the Equivalence of Breaking the Diffie-Hellman Protocol and Computing Discrete Logarithms*

Remember: El Gamal Encryption

- Public key Alice is (G, g, g^a)
- Private key Alice is a
- (Probabilistic) encryption of $m \in G$ is (c_1, c_2) where
  - $c_1 = g^b$ for some random b
  - $c_2 = mg^{ab}$
- Alice can decrypt (c_1, c_2) with private key
  - Compute $c_1^a = g^{ab}$
  - Compute $c_2/g^{ab} = m$

Static Diffie-Hellman

- Usual Diffie-Hellman oracle: given **random** g^x and g^y, return g^{xy}
- **Static** Diffie-Hellman oracle: given g^x, returns g^{ax} **for some fixed a** (less powerful than DH oracle)
- Can be built using ElGamal decryption oracle
  - Send (g^x, c_2) for arbitrary c_2 to decryption oracle, get $m = c_2/g^{ax}$, and deduce $g^{ax} = c_2/m$

  (remember IND-CCA security definition: adversary may query decryptions of their choice)
- Does this oracle make DLP (g, g^a) easier?

Static Diffie-Hellman

- If $|G| - 1 = uv$, can solve DLP in time $u + \sqrt{v}$ with u calls to the static Diffie-Hellman oracle
  - If $v \approx u^2$ this gives a $O(|G|^{1/3})$ DLP algorithm with $O(|G|^{1/3})$ calls to the oracle
- Idea:
  - Use SDH oracle to produce g^{x^u}
  - Let $x = g_0^{x'}$, with g_0 generator of $\mathbb{Z}^*_{|G|-1}$
  - x^u lies in order v multiplicative subgroup
  - Get x' mod v with "Baby Step Giant Step in exponent"
  - Reduce to a DLP in subgroup of order u, which is solved again with BSGS
- See Brown-Gallant, *The Static Diffie-Hellman Problem*

DLP vs DDH

- There exist groups where DLP is believed to be hard but DDH is easy
- Indeed for elliptic curve groups equipped with an efficient symmetric pairing, we have

$$(P, aP, bP, cP) \text{ is a DDH tuple} \Leftrightarrow e(P, cP) = e(aP, bP)$$

Other "DLP variants'

- The cryptography literature is rich of variants of DLP/CDH problems
- Purpose is provable security: a protocol cannot be proved secure based on DLP only, so a variant is used instead
- Of course:
  - This only makes sense if variant is indeed hard
  - Confidence in hardness only gained by cryptanalysis
- Standard practice: show the variant is hard in the "generic group model"
- See Koblitz-Menezes papers ...

Factorization vs Discrete logarithms

- Discrete logarithm and factorization problems may seem unrelated, yet algorithms to solve them are similar
- Exception (?): quasi-polynomial time algorithm for discrete logarithms in small to medium characteristic
- Hardness of large characteristic field discrete logarithms and integer factorization is comparable today

Pollard's rho

- Factorization problem: given $n = pq$, find p and q
- Idea: find x and y such that $\gcd(x - y, n) = p$
  in other words $x = y \bmod p$ but $x \neq y \bmod n$
- Define some "pseudorandom" iteration function f
- Compute iterates x_i and x_{2i}
- Simultaneously compute $\gcd(x_i - x_{2i}, n)$
- By birthday's paradox,
  $x_i = x_{2i} \bmod p$ after $O(p^{1/2})$ steps on average,
  $x_i = x_{2i} \bmod n$ after $O(n^{1/2}) = O(p)$ steps on average
- Hence we succeed after $O(p^{1/2})$ steps on average

Pollard's $p-1$ method

- Factorization problem: given $n = pq$, find p and q
- A number $x = \prod p_i^{e_i}$ is B-powersmooth if $p_i^{e_i} < B$
- The method assumes that $p-1$ is B-powersmooth, and $q-1$ is not
- Let s be the product of all $p_i^{e_i} < B$
- By assumption $(p-1)|s$, hence $g^s = 1 \bmod p$ (unless $\gcd(g,p) = p$ in which case we deduce p)
- We deduce $p \mid \gcd(g^s - 1, n)$
- First step in gcd computation is $g^s - 1 \bmod n$, computed using square-and-multiply algorithm
- Only works if some factor p such that $p-1$ smooth; elliptic curve method (ECM) removes this constraint

Other factorization methods

- Quadratic field sieve $L_n(1/2)$
- Elliptic curve factorization method $L_p(1/2)$
- (General) number field sieve $L_n(1/3)$

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Other cryptanalysis tools

- Above slides only cover a sample of cryptanalysis tools
- Many more cool algorithms to tackle the core computational problems
  - Coppersmith's techniques
  - Groebner basis algorithms
  - Function field sieve
  - Multicollision attacks
  - Gentry-Szydlo's algorithm
  - . . .
- In practice, security often relies on variants easier to solve

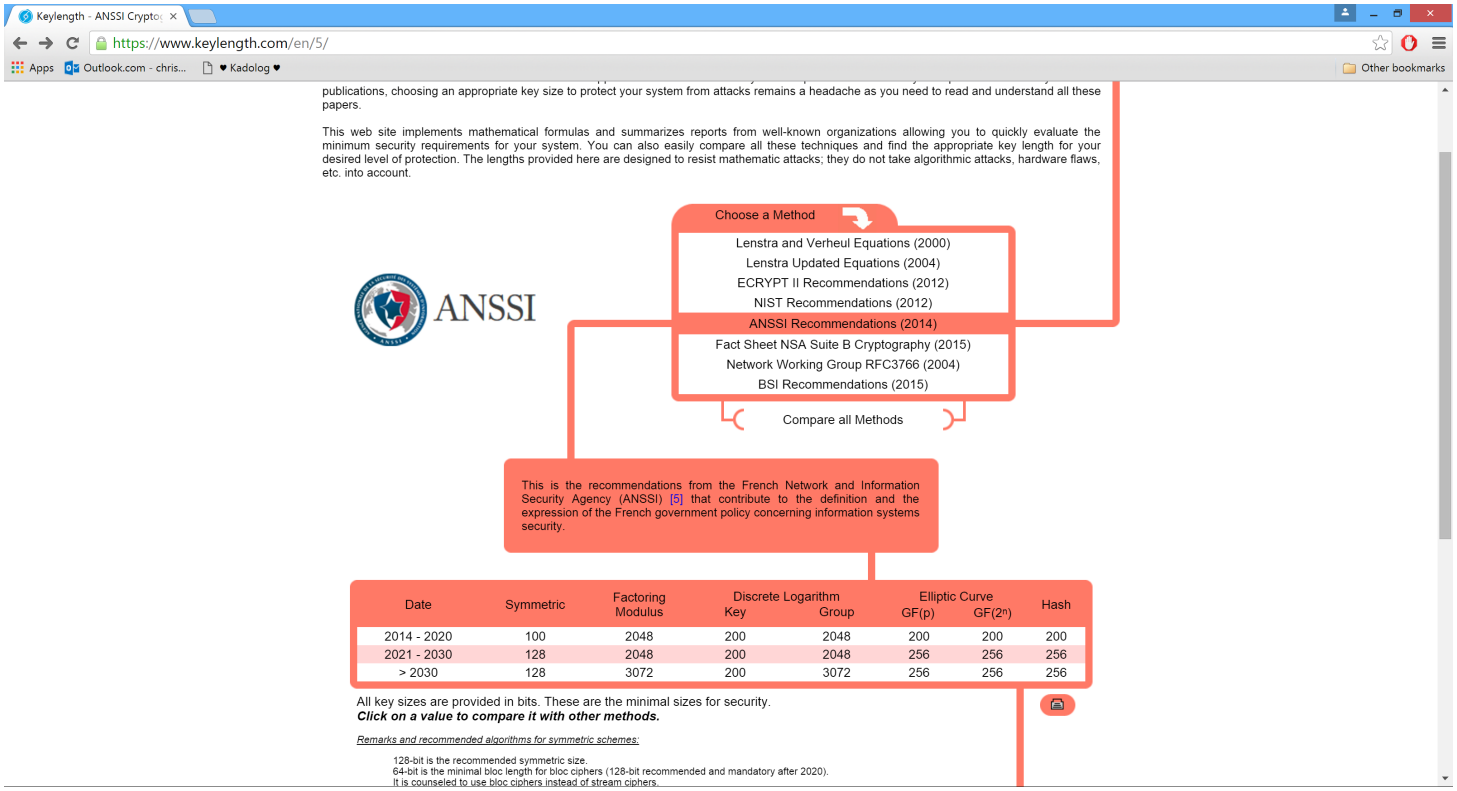
Selecting cryptographic key lengths

- Reasonably easy for symmetric key cryptography, much harder for public key cryptography
- Efficiency vs security tradeof
- Understand complexity of best cryptanalysis algorithms, both asymptotically and in practice
- Tailor key lengths to your needs
  - Short-term or long-term security?
  - Facing the NSA or your little brother?
- Account for Moore's law and algorithmic progress
- Take some security margin
- See Lenstra-Verheul, *Selecting Cryptographic Key Sizes*

Public challenges

- Goal: encourage cryptanalysis attempts to estimate concrete hardness of a problem, and parameters to be used in practice
  - RSA factoring challenges (ended 2007)
  - Certicom ECDLP challenges (solved up to 109 bits)
  - Darmstadt lattice challenges
  - Fukuoka MQ challenges
- Contribute and enter "Hall of Fame" of any of them!

Recommended key lengths



- Check `www.keylength.com` for updates!

Conclusion

- Cryptanalysis attempts are essential for cryptography
- Cryptanalysis relies on a wide range of mathematical tools
- DLP algorithms much more efficient for finite fields than elliptic curves
- Many (large characteristic) DLP algorithms have an analogous factoring algorithm
- Beware of "problem variants" that have been little (if at all) studied